EXP 14: Implement House price Prediction using appropriate machine learning algorithm

CODING:

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# House Price Prediction using Linear Regression

# Training dataset
# Area in square feet
area = [1000, 1200, 1500, 1800, 2000]

# House prices in rupees
price = [2000000, 2400000, 3000000, 3600000, 4000000]

n = len(area)

# Calculate mean
mean_x = sum(area) / n
mean_y = sum(price) / n

# Calculate slope
numerator = 0
denominator = 0

for i in range(n):
    numerator += (area[i] - mean_x) * (price[i] - mean_y)
    denominator += (area[i] - mean_x) ** 2

slope = numerator / denominator
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# Calculate intercept
intercept = mean_y - slope * mean_x

# Predict house price
new_area = 1600
predicted_price = slope * new_area + intercept

print("HOUSE PRICE PREDICTION")
print("----------------------")
print("House Area:", new_area, "sq.ft")
print("Predicted House Price: Rs.", round(predicted_price))
```

OUTPUT:

